University of Debrecen
Faculty of Engineering
Department of Electrical Engineering and Mechatronics

Robotic Manipulator Arm Design Report

Integrated Mechatronic System Design

2024/25/II. semester

Title:	Robotic Manipulator Arm Design with Integrated Motor Control
Name:	Shahd Negm
Neptun:	
Department:	Mechatronics
Date:	June 14, 2025
Signature:	Shahd Negm

Contents

1 Introduction **2**

2 Design Objectives **2**

3 Design Process and Thought Development **2**
3.1 Base and First Joint 2
3.2 Middle Linkages 3
3.3 Top Joint and End-Effector 3

4 Mechanical Design and Assembly **3**

5 Motor and Encoder Specifications **4**

6 Custom PCB Motor Control System **4**
6.1 Overview 4
6.2 Main Components 4
6.3 Wiring Diagram 5

7 Cyber-Physical System Integration **6**

8 Electrical Machines and Drives Overview **6**
8.1 Simulation Setup 6
8.2 Control Strategies Implemented 6
8.3 Simulink Results 7
8.4 Relevance to Robot Application 7

9 Applications **7**

10 Conclusion **7**

1 Introduction

This report presents the development of a 4-DOF robotic manipulator arm for light industrial drilling. The system integrates structural design, embedded control, and simulation compatibility. Each component was designed in alignment with the selected motor's dimensions and mechanical properties to ensure practical feasibility.

2 Design Objectives

- Mechanically integrate four Maxon DCX 22 S motors into a robotic arm.
- Enable precision drilling through stable torque and encoder feedback.
- Design a PCB motor controller capable of driving high stall currents.
- Simulate the robot in MATLAB/Simulink using proper joint constraints.
- Create a modular design with clear cable routing and standard fasteners.

3 Design Process and Thought Development

The design began with selecting a motor that matched both the torque requirement and compact mounting needs. After reviewing several options, the Maxon DCX 22 S was selected.

3.1 Base and First Joint

The base acts as a mechanical shield around the motor. Two face-mounting holes on the motor were aligned using extruded features. The first joint provides 360° rotation and houses the wiring cavity for safe routing.

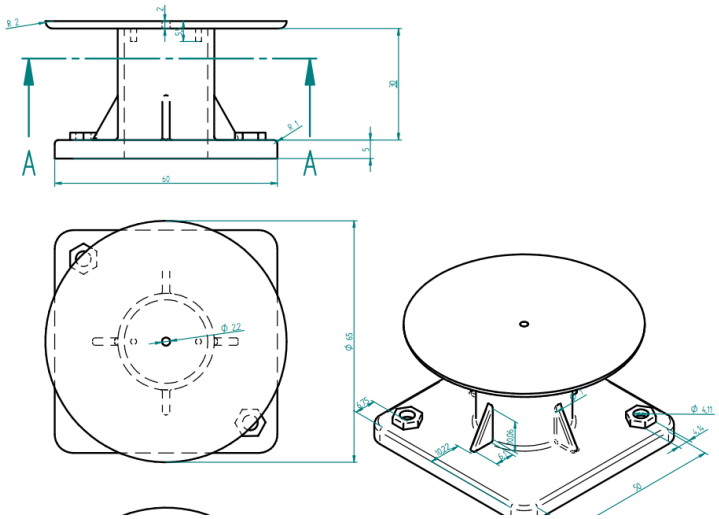


Figure 1: CAD model of the base housing the first motor

3.2 Middle Linkages

Two arm segments are driven by a second motor placed between them. A revolute joint at the center allows vertical lifting while limiting backward motion to avoid collisions. These linkages were modeled using stainless steel profiles for strength.

3.3 Top Joint and End-Effector

A third constrained revolute joint allows the arm to position the tool in variable angles while respecting link limitations. The final joint adds mobility to the end-effector for better reach and to avoid singularities during planar motion.

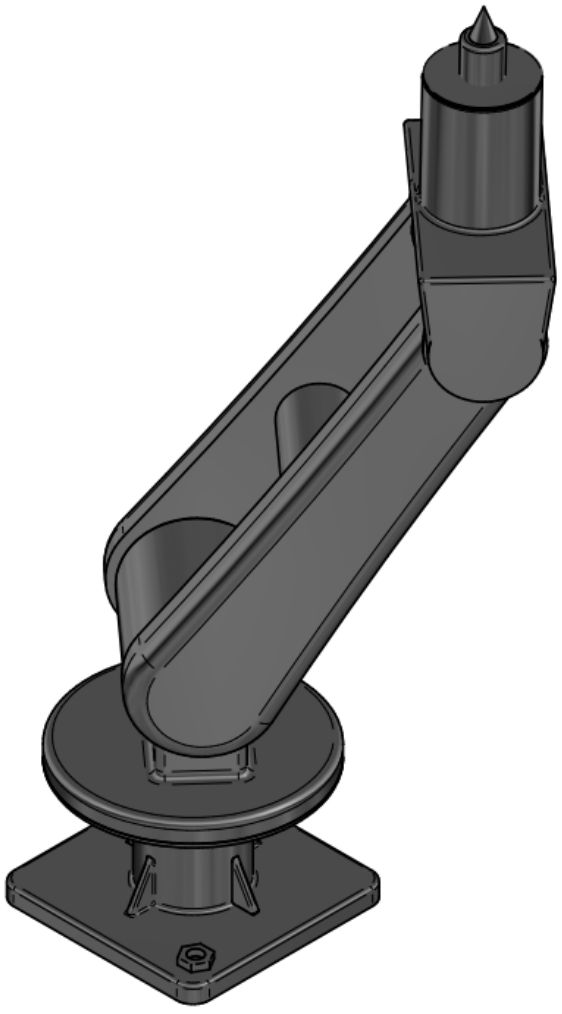


Figure 2: Final assembly of the 4-DOF robotic manipulator

4 Mechanical Design and Assembly

- All components are made of stainless steel (304, 316, and 420).
- Joints are connected using shafts and M3 screws for accessibility.
- Cable channels are built into the motor housings for clean routing.

- All technical drawings were generated and dimensioned in Solid Edge.

5 Motor and Encoder Specifications

Motor: Maxon DCX 22 S

- Voltage: 12V
- Stall Torque: 108 mNm
- Nominal Speed: 10,700 RPM
- Resistance: 1.02 Ω

Encoder: Maxon ENX 10 EASY

- Type: Incremental optical encoder
- Resolution: Configurable
- Interface: Digital signal compatible with microcontrollers

6 Custom PCB Motor Control System

6.1 Overview

The PCB design includes full H-bridge control using VNH5019A-E and supports dual power sources. Key design principles included current isolation, voltage regulation, and debugging interfaces.

6.2 Main Components

- **Microcontroller:** ATmega328P
- **Motor Driver:** VNH5019A-E (high-current)
- **Power Selection:** USB or external via switch
- **Voltage Regulation:** 12V to 5V logic
- **Interfaces:** UART, ISP header, digital/analog breakout
- **Debugging:** Status LEDs on motor and logic lines

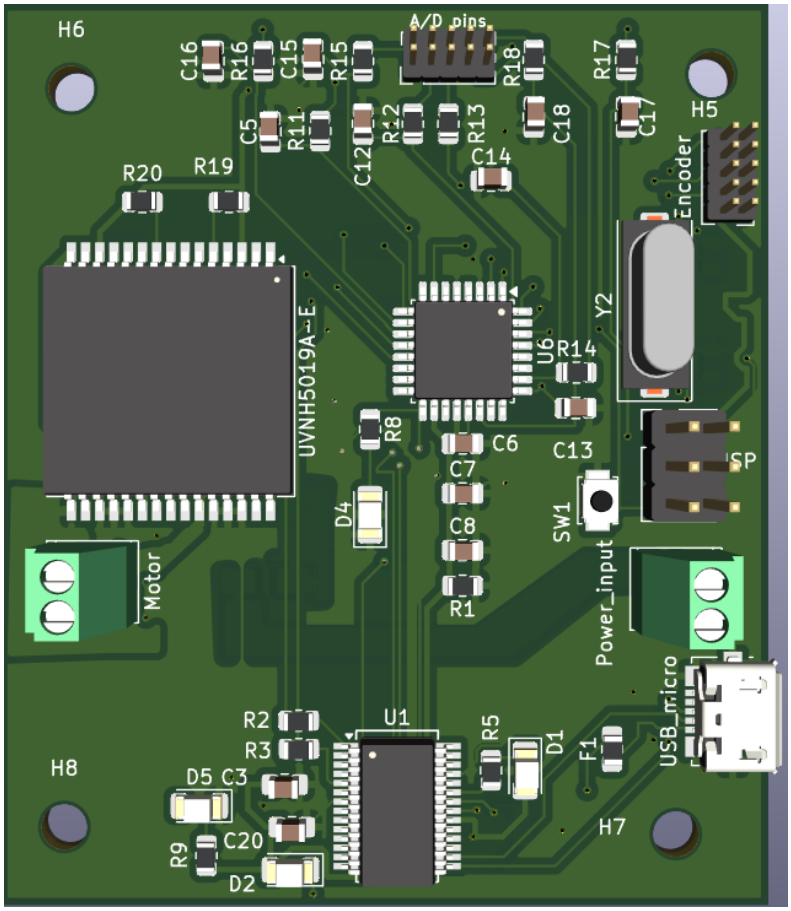


Figure 3: Top view of the custom PCB designed in KiCAD

6.3 Wiring Diagram

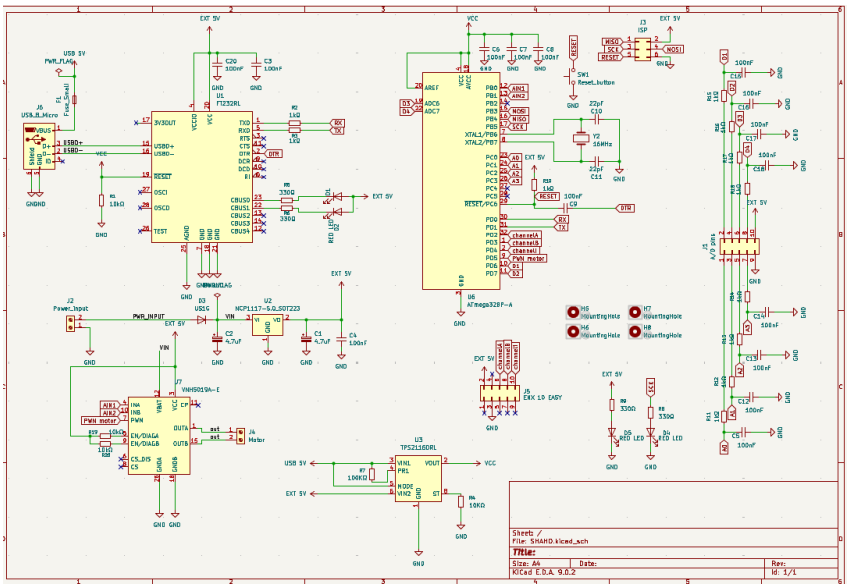


Figure 4: Wiring diagram showing connection to motor and encoder

7 Cyber-Physical System Integration

The robotic manipulator was prepared for simulation and control validation using MATLAB and Simulink's Simscape Multibody environment. The integration process included the following steps:

- The full CAD model was exported from Onshape with each joint explicitly defined and tagged.
- The model was imported into Simulink using the Simscape Multibody Converter, preserving the 3D geometry and assembly structure.
- Joint types (revolute, rigid) were mapped accordingly in Simulink.
- Joint limits were added in the MATLAB code to reflect real mechanical constraints, preventing unrealistic rotation beyond physical range.
- Forward kinematics were calculated using Denavit-Hartenberg parameters for position verification.
- Inverse kinematics functions were used to determine required joint angles for target positions.
- The model is now ready for further control development, including motion planning and feedback integration.

8 Electrical Machines and Drives Overview

8.1 Simulation Setup

The Maxon DCX 22 S motor used in the robotic arm was modeled and tested in MATLAB/Simulink as part of the Electrical Machines and Drives course. The simulations aimed to evaluate different control strategies suitable for precision tasks like drilling.

8.2 Control Strategies Implemented

Three primary control methods were explored:

- **PI Control:** A manually tuned proportional-integral controller to manage motor speed.
- **PID Control:** A position control system with derivative tuning for reduced overshoot.
- **Cascade Control:** A combined current-speed loop to improve accuracy and stability.

8.3 Simulink Results

The speed and position controllers were validated using simulation blocks and step inputs. The cascade controller showed the best performance for tasks requiring torque regulation, such as drilling.

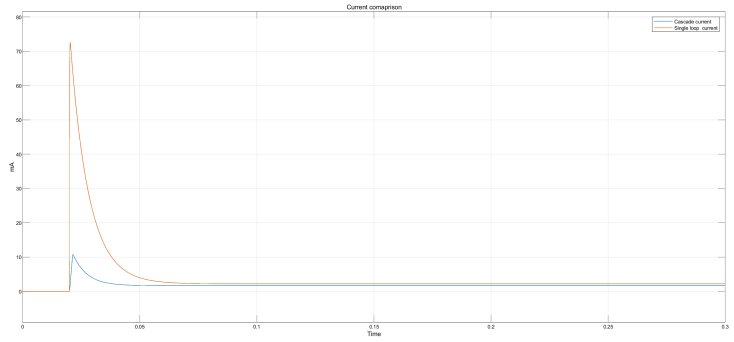


Figure 5: current response with cascade control implementation

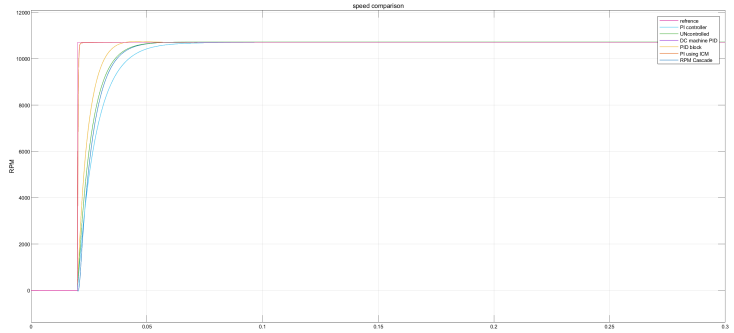


Figure 6: Position tracking performance under PID control

8.4 Relevance to Robot Application

The tested control loops are directly applicable to the designed manipulator, especially the cascade control, which ensures current limiting and smoother operation during high-load tasks like drilling.

9 Applications

This robot is capable of:

- PCB or plastic drilling with controlled torque

10 Conclusion

This 4-DOF robotic arm was designed from the motor outward. Each joint's geometry was defined by the mounting requirements of the Maxon DCX 22 S and its encoder. A fully functional PCB was created to control the motors, and MATLAB simulation compatibility ensures future development of automated routines. The robotic system is mechanically feasible, electronically safe, and academically robust.